

Exchange Rates and Best Buys



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Exchange rates: multiply one way, divide the other. Best buys: same unit, same currency, then compare and say which.

THE TOPIC ON ONE MAP

Exchange Rates and Best Buys

01 — Converting currencies

\$306 into euros at $\text{€}1 = \$1.125$

100-koruna notes

conversion graph

02 — Compare prices in two currencies

camera in Hong Kong

boots in London, Geneva, Paris

petrol UK vs USA

03 — Best buys

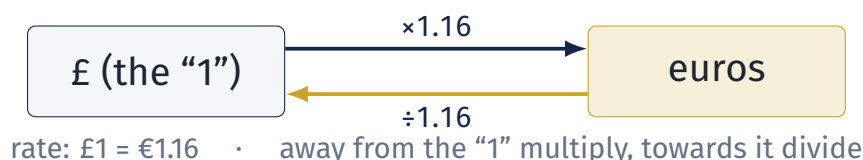
trays of plants

crisps packs

discount offers

water meter or not

MULTIPLY OR DIVIDE? Read the rate as “1 of these = so many of those”:



01 Converting Currencies

Find the side of the rate that says “1”. Going **away** from it: multiply. Going **towards** it: divide. Check the size makes sense.

WHEN YOU SEE → YOU DO

“£1 = 25.82 koruna ... change £200”

→ away from the 1: $\times 25.82$

“€1 = \$1.125 ... change \$306 into euros”

→ towards the 1: $\div 1.125$

“only 100-koruna notes, as many as possible”

→ round **down** to whole notes

“complete: \$1 = ... Namibian dollars”

→ divide: $3891 \div 300$

a conversion graph

→ read across and down; for big amounts scale up

WORKED MODEL • TOWARDS THE 1

Adele changes \$306 into euros. €1 = \$1.125.

1 dollars → euros is towards the “1”

decide

2 $306 \div 1.125$

divide

ANSWER €272

× **TRAP** Multiplying the wrong way gives €344 — more euros than dollars, though a euro is worth more. Always check the size.

IDEA 01

Practise: Converting Currencies

My questions first, then selected practice.

Q1 **DR ESLAM** Away from the 1

CH 1 • TOPIC 14 2 marks

£1 = 5.2 dirhams. Change £350 into dirhams.

Q2 **DR ESLAM** Towards the 1

CH 1 • TOPIC 14 2 marks

£1 = \$1.25. Change \$450 into pounds.

SELECTED PRACTICE

Q3

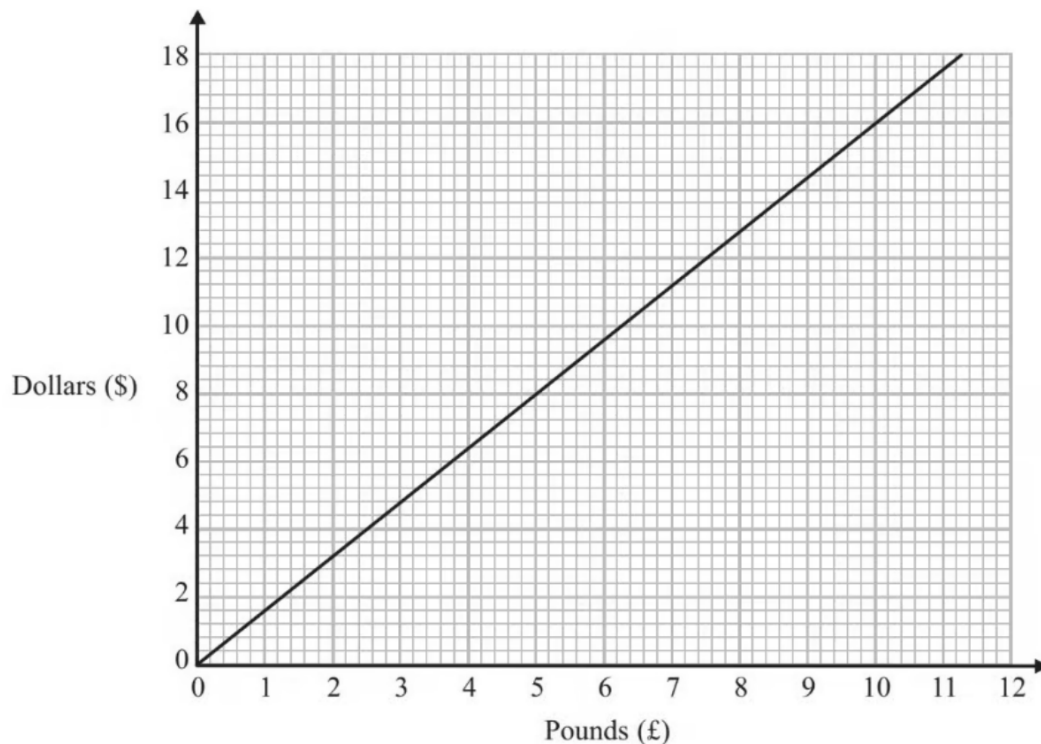
PRACTICE

Practice • Easy

CH 1 • TOPIC 14

5 marks

6 (a) You can use this conversion graph to change between pounds (£) and dollars (\$).



Use the conversion graph to change £5 to dollars.

(1 mark)

(b) Ella has \$200 and £800

Her hotel bill is \$600

Ella pays the bill with the \$200 and some of the pounds.

Use the conversion graph to work out how many pounds she has left.

(4 marks)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 14 2 marks

- 8 Rashid changes 30 000 rupees to dollars when the exchange rate is \$1 = 68.14 rupees.

How many dollars does he receive?

\$

(2 marks)

Q5**PRACTICE**

Practice • Easy

CH 1 • TOPIC 14 2 marks

- 10 Adele changes \$306 into euros (€) when the exchange rate is €1 = \$1.125.

Calculate the number of euros she receives.

€

(2 marks)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 14 5 marks

7 (a) Asif is going on holiday to Turkey.

The exchange rate is $\text{£}1 = 3.5601$ lira.

Asif changes $\text{£}550$ to lira.

Work out how many lira he should get.

Give your answer to the nearest lira.

(2 marks)

(b) Asif sees a pair of shoes in Turkey. The shoes cost 210 lira.

Asif does not have a calculator.

He uses $\text{£}2 = 7$ lira to work out the approximate cost of the shoes in pounds.

Use $\text{£}2 = 7$ lira to show that the approximate cost of the shoes is $\text{£}60$

(2 marks)

(c) Is using $\text{£}2 = 7$ lira instead of using $\text{£}1 = 3.5601$ lira a sensible start to Asif's method to work out the cost of the shoes in pounds?

You must give a reason for your answer.

(1 mark)

Q7**PRACTICE**

Practice · Medium

CH 1 · TOPIC 14 2 marks

- 9 A new model locomotive costs \$64.

Calculate the cost of the locomotive in rupees when the exchange rate is 1 rupee = \$0.0154.
Give your answer correct to the nearest 10 rupees.

..... rupees

(2 marks)

Q8**PRACTICE**

Practice · Medium

CH 1 · TOPIC 14 2 marks

- 11 The Muller family are on holiday in New Zealand.

They change some euros (€) and receive \$1962 (New Zealand dollars).
The exchange rate is €1 = \$1.635 .

Calculate the number of euros they change.

€

(2 marks)

02 Compare Prices in Two Currencies

Convert everything into **one** currency (usually pounds), and into the **same unit** (per litre, per kg). Then compare and give the difference.

WHEN YOU SEE → YOU DO

“how much cheaper in Hong Kong?”

→ convert, subtract

“London, Geneva or Paris?”

→ all three into pounds

“per US gallon ... 1 gallon = 3.79 litres”

→ price per litre in the same currency

“give your answer in dollars, to the nearest cent”

→ convert the other price into dollars

Same currency

Same unit

Compare

Say which and by how much

WORKED MODEL • PETROL

UK: £1.24 per litre. USA: \$3.15 per US gallon. 1 gallon = 3.79 L, £1 = \$1.47.
Which is cheaper?

1 $3.15 \div 3.79 = \$0.831$ per litre

same unit

2 $0.831 \div 1.47 = £0.565$ per litre

same currency

ANSWER USA is cheaper (£0.57 < £1.24)

IDEA 02

Practise: Compare Prices in Two Currencies

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM Compare

CH 1 • TOPIC 14 3 marks

A watch costs £84 in London and €99 in Rome. £1 = €1.15. Where is it cheaper, and by how much?

SELECTED PRACTICE**Q10** PRACTICE Practice • Medium

CH 1 • TOPIC 14 3 marks

4 Ben goes on holiday to Hong Kong.

In Hong Kong, Ben sees a camera costing HK\$3179.55

In London, an identical camera costs £285

The exchange rate is £1 = HK\$12.30

Ben buys the camera in Hong Kong.

How much cheaper is the camera in Hong Kong than in London?

(3 marks)

Q11**PRACTICE**

Practice • Medium

CH 1 • TOPIC 14 **2 marks**

12 A dressmaker charges \$35 or 2300 rupees to make a dress.

Calculate the difference in price when the exchange rate is 1 rupee = \$0.0153 .

Give your answer in rupees.

..... rupees

(2 marks)

Q12**PRACTICE**

Practice • Hard

CH 1 • TOPIC 14

3 marks

4 Identical pairs of boots are sold in London, in Geneva and in Paris.

These boots have a price of

£115 in London

189 Swiss francs in Geneva

174 euros in Paris

The exchange rates are

£1 = 1.39 Swiss francs

£1 = 1.27 euros

Are the boots the best value for money in London or in Geneva or in Paris?

You must show how you get your answer.

(3 marks)

Q13**PRACTICE**

Practice · Hard

CH 1 • TOPIC 14 **3 marks**

- 10** On holiday Maria paid 2.25 euros for a newspaper when the exchange rate was \$1 = 0.9416 euros.
At home Maria paid \$1.13 for the newspaper.

Calculate the difference in price.

Give your answer in dollars, correct to the nearest cent.

\$

(3 marks)

PAST PAPERS

Q14 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 2H · Q12

CH 1 · TOPIC 14 4 marks

- 12 Astrid wants to buy some oil.
She can buy the oil from either Dane Oil or Arctic Oil.

Here is information about the price that each company will charge Astrid.

Dane Oil	Arctic Oil
(4.2×10^5) litres for 2 500 000 Krone	(8.6×10^5) litres for 770 000 Dollars

Astrid wants to get the better value for money for the oil.

$$1 \text{ Dollar} = 6.57 \text{ Krone}$$

From which company should she buy her oil, Dane Oil or Arctic Oil?
You must show your working.

(Total for Question 12 is 4 marks)

03 Best Buys

Work out the cost of **one unit** (per plant, per ml, per bag) for every option. The smallest cost per unit is the best value.

WHEN YOU SEE → YOU DO

“which size is the best value for money?”

→ price ÷ quantity for each

“60% off, then 15% off” vs “ $\frac{2}{3}$ off”

→ final price of each

“buy 2, get 3 more at half price”

→ price of exactly the number needed

“should he have a water meter?”

→ yearly cost of each option

ZONE COST PER UNIT

price ÷ amount

smallest wins

ZONE AMOUNT PER £

amount ÷ price

biggest wins

WORKED MODEL • COFFEE

12 sachets £5.65, 20 sachets £9.20, 35 sachets £15.75. Best value?

1 47.1p, 46p, 45p per sachet

price ÷ sachets

ANSWER the large box (45p per sachet)

× **TRAP** Comparing the prices themselves. The large box costs most but is cheapest per sachet.

IDEA 03

Practise: Best Buys

My questions first, then selected practice.

Q15 **DR ESLAM** Best buy

CH 1 • TOPIC 14 3 marks

500 ml for £1.40 or 750 ml for £2.05. Which is the better value?

SELECTED PRACTICE**Q16** **PRACTICE** Practice • Easy

CH 1 • TOPIC 14 4 marks

- 1 Plants are sold in three different sizes of tray.

A small tray of 30 plants costs £6.50

A medium tray of 40 plants costs £8.95

A large tray of 50 plants costs £10.99

Kaz wants to buy the tray of plants that is the best value for money.

Which size tray of plants should she buy?

You must show all your working.

(4 marks)

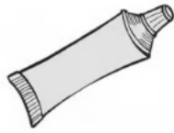
Q17**PRACTICE**

Practice · Easy

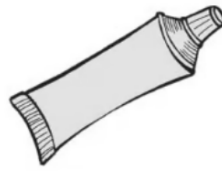
CH 1 · TOPIC 14

4 marks

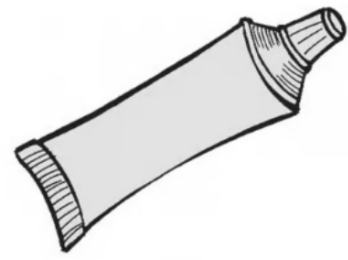
3 Toothpaste is sold in three different sizes of tube.



50ml



75ml



125ml

A 50 ml tube costs £ 1.09

A 75 ml tube costs £ 1.68

A 125 ml tube costs £ 2.69

Which tube of toothpaste is the best value for money?

You must show all your working.

(4 marks)

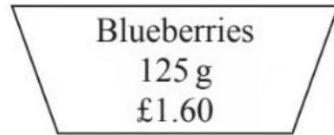
Q18**PRACTICE**

Practice · Easy

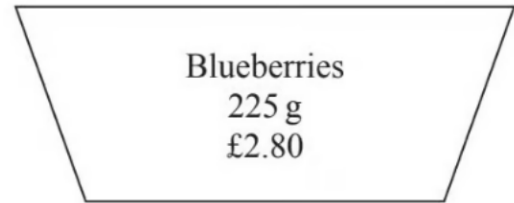
CH 1 · TOPIC 14

3 marks

5 Blueberries are sold in small cartons and in large cartons.



small carton



large carton

There are 125 g of blueberries in a small carton.

Each small carton costs £1.60

There are 225 g of blueberries in a large carton.

Each large carton costs £2.80

Which size of carton is the better value for money? You must show your working.

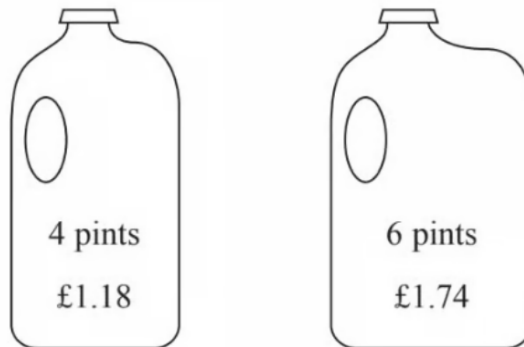
(3 marks)

Q19**PRACTICE**

Practice • Medium

CH 1 • TOPIC 14 3 marks

2 Milk is sold in two sizes of bottle.



A 4 pint bottle of milk costs £1 .18

A 6 pint bottle of milk costs £1 .74

Which bottle of milk is the best value for money?

You must show all your working.

(3 marks)

Q20**PRACTICE**

Practice • Hard

CH 1 • TOPIC 14 5 marks

- 2 Henry is thinking of having a water meter.

These are the two ways he can pay for the water he uses.

Water Meter

A charge of £28.20 per year

plus

91.22p for every cubic metre of water used

1 cubic metre = 1000 litres

No Water Meter

A charge of £107 per year

Henry uses an average of 180 litres of water each day.

Use this information to determine whether or not Henry should have a water meter.

(5 marks)

Q21**PRACTICE**

Practice · Hard

CH 1 · TOPIC 14 3 marks

9 Bottles of drink are for sale at three shops.

The normal price of a bottle is the same at each shop.

Shop A Buy 1 bottle Get 2 more bottles at half price	Shop B Buy 2 bottles Get 3 more bottles at half price
Shop C 30% off a bottle	

What is the cheapest way to buy **exactly** 8 bottles?

You can buy from more than one shop.

You **must** show your working.

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------|---|
| 01 | Convert | Away from the “1”: multiply. Towards it: divide.
Check the size. |
| 02 | Compare | Same currency, same unit, then the difference. |
| 03 | Best buy | Cost per unit — smallest wins; say which. |

BOTTOM LINE Find the “1” → multiply away, divide towards → compare like with like.

Answers

Exchange Rates and Best Buys • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • CONVERTING CURRENCIES

Q1	Dr Eslam • Away from the 1	$350 \times 5.2 = 1820$ dirhams
Q2	Dr Eslam • Towards the 1	$450 \div 1.25 = £360$
Q3	Practice • Easy	(a) \$7.50 (b) \$400 = £267: about £533 left
Q4	Practice • Easy	\$440.27
Q5	Practice • Easy	€272
Q6	Practice • Medium	(a) 1958 lira (b) $210 \div 7 \times 2 = 60$ (c) yes: $3.5 \approx 3.56$
Q7	Practice • Medium	4160 rupees
Q8	Practice • Medium	€1200

IDEA 02 • COMPARE PRICES IN TWO CURRENCIES

Q9	Dr Eslam • Compare	Rome: $99 \div 1.15 = £86.09$: London cheaper by £2.09
Q10	Practice • Medium	£258.50: £26.50 cheaper
Q11	Practice • Medium	12 rupees
Q12	Practice • Hard	Geneva £135.97, Paris £137.01: London (£115)
Q13	Practice • Hard	\$2.39 abroad: \$1.26 more
Q14	Edexcel 4MA1 • January 2020 • Paper 2H • Q12	Dane \$0.906 per litre, Arctic \$0.895: Arctic Oil

IDEA 03 • BEST BUYS

Q15	Dr Eslam • Best buy	0.28p vs 0.273p per ml: the 750 ml bottle
Q16	Practice • Easy	21.7p, 22.4p, 22.0p per plant: small tray
Q17	Practice • Easy	2.18p, 2.24p, 2.15p per ml: 125 ml
Q18	Practice • Easy	1.28p vs 1.24p per g: large carton
Q19	Practice • Medium	29.5p vs 29p per pint: 6 pints
Q20	Practice • Hard	meter £88.13 < £107: yes, have a meter
Q21	Practice • Hard	Shop A twice (6 bottles for 4) + 2 from Shop C: 5.4 bottle-prices